Towards a more inductive world: A review on graph embedding methods for drug repurposing approaches

J. de la Fuente[1,2], G. Serrano[2], U. Veleiro[2,4], M. Casals[1], O. Azurmendi-Senar[2,3], A. Pineda-Lucena[4], I. Ochoa[1], S. Vicent[3], O. Gevaert[5], M. Hernaez[2]

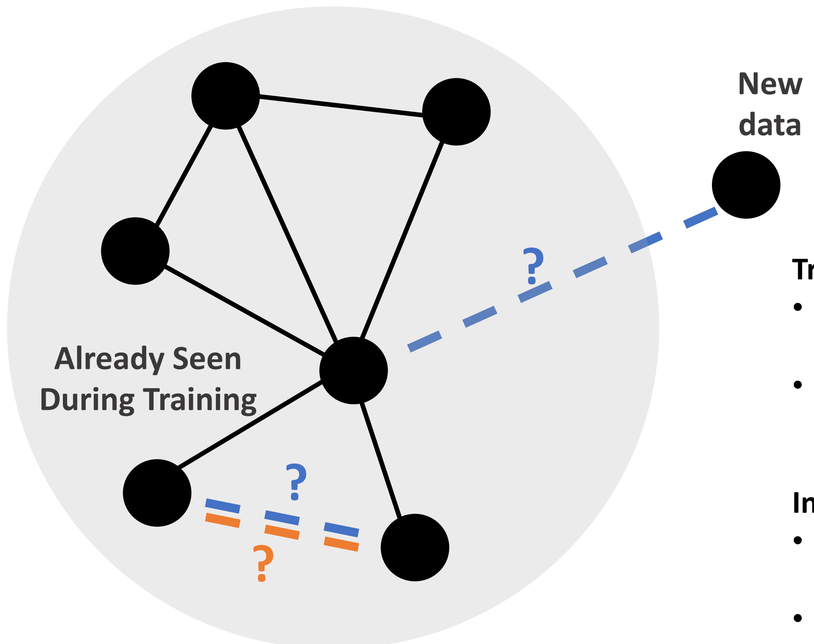
[1]Tecnun University of Navarra, [2]Computational Biology Program, CIMA University of Navarra, [3]Solid Tumors Program, CIMA University of Navarra, [4]Molecular Therapeutics Program, CIMA University of Navarra, [5]Standford Center for Biomedical Informatics Research (BMIR), Stanford University.

Background

The core of drug discovery is to find a potential candidate for a given pharmacological target responsible for a certain disease. The need of reducing the time required to predict those **Drug Target Interactions (DTIs)** has fueled the development of algorithms, especially in drug repurposing projects, where the finding of new uses for previously approved drugs inherently speeds up their entry into the market.

Motivation

Several **machine learning methods** have been proposed **to predict DTIs considering** complex heterogeneous **graphs**. These techniques, which generate embeddings from graph nodes, can be classified as inductive or transductive. While the former store information during training to generate embeddings for unseen nodes, the latter only generate embeddings for nodes seen during training leading to data leakage at the evaluation stage. Thus, it is of utmost **importance to propose fair and generalizable evaluation guidelines**. Besides, current ***golden standard*** **datasets are small and no longer representing reality.**



Transductive vs Inductive

Transductive learning:

- Builds a model that fits training and testing data points allowing to predict unlabeled edges.
- Can lead to data leakage at the evaluation stage.

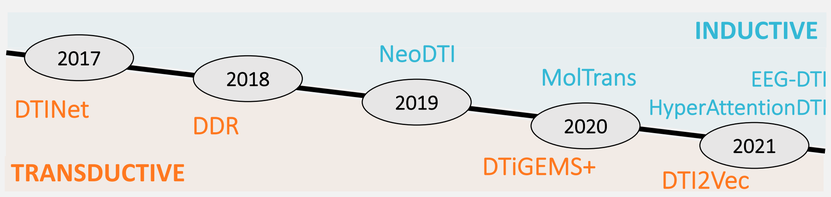
Inductive learning:

- Builds a generic predictive model. If new nodes are added, we can still use the trained model.
- Train the model and predict interaction between nodes that we have never encountered before

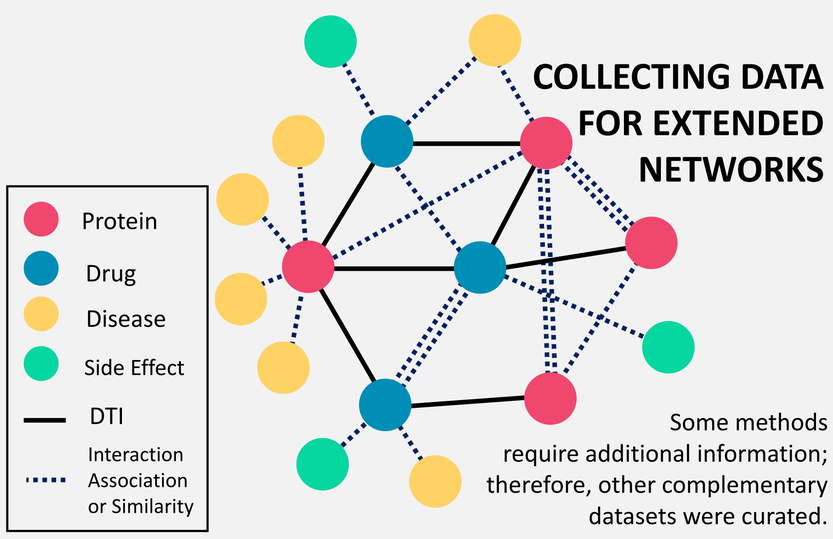
MATERIALS & METHODS

MODELS AND DATASETS

- 8 Models (4 transductive, 4 inductive)
- 8 DTI Datasets commonly used (Yamanishi, BindingDB, DAVIS, DrugBank)



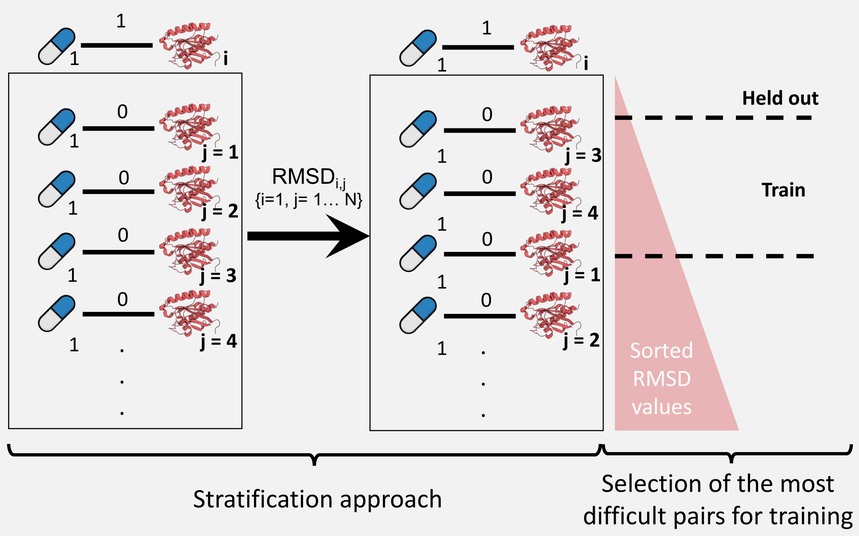
COLLECTING DATA FOR EXTENDED NETWORKS



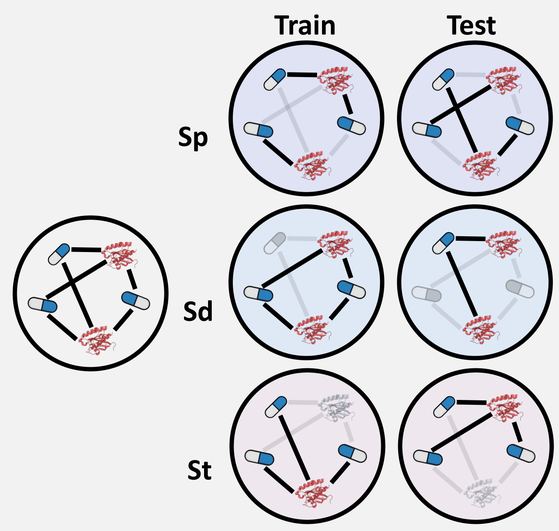
Some methods require additional information; therefore, other complementary datasets were curated.

CONSIDERING OTHER CRITERIA FOR NEGATIVE SUBSAMPLING

- **DTI matrices are sparse** and negative edges come from combinatory. **DTI data is imbalanced** since non-positive labels are far more abundant than positive ones.
- Root Mean Square Deviation (**RMSD**) of protein structures was calculated, as protein structure is more conserved than sequence itself. **The smaller** the RMSD between two proteins, **the similar** they are.
- **Idea**: **Make it difficult to the model**, train with not too similar proteins.



A FAIR EVALUATION SCHEME: ALTERNATIVE SPLITTING DATA



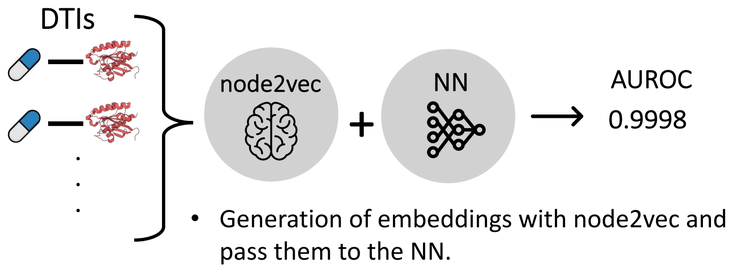
It is not trivial to split the network in two, as a train/test set, while preserving network characteristics.

It has been recently proposed the following evaluation scheme:

- **Sp**. Any protein or drug node may appear both in the train and test set, but interactions (edges) cannot.
- **Sd**. Drug nodes are not duplicated in the train and test set.
- **St**. Protein nodes are not duplicated in train and test set.

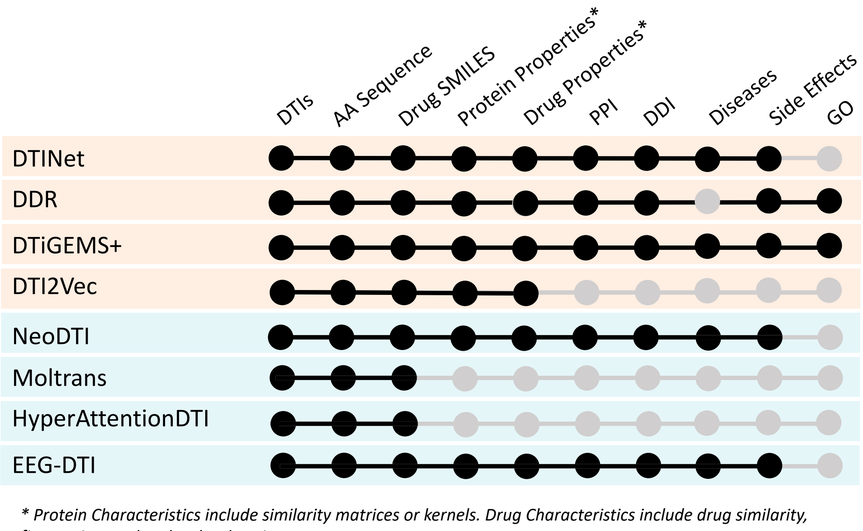
RESULTS

TRANSDUCTIVITY LEADS TO OVER-SIMPLISTIC INFERRING AND OVERFITTING



- Generation of embeddings with node2vec and pass them to the NN.
- Node2vec learns strictly from network topology.
- AUROCs indicate that the model is overfitting.

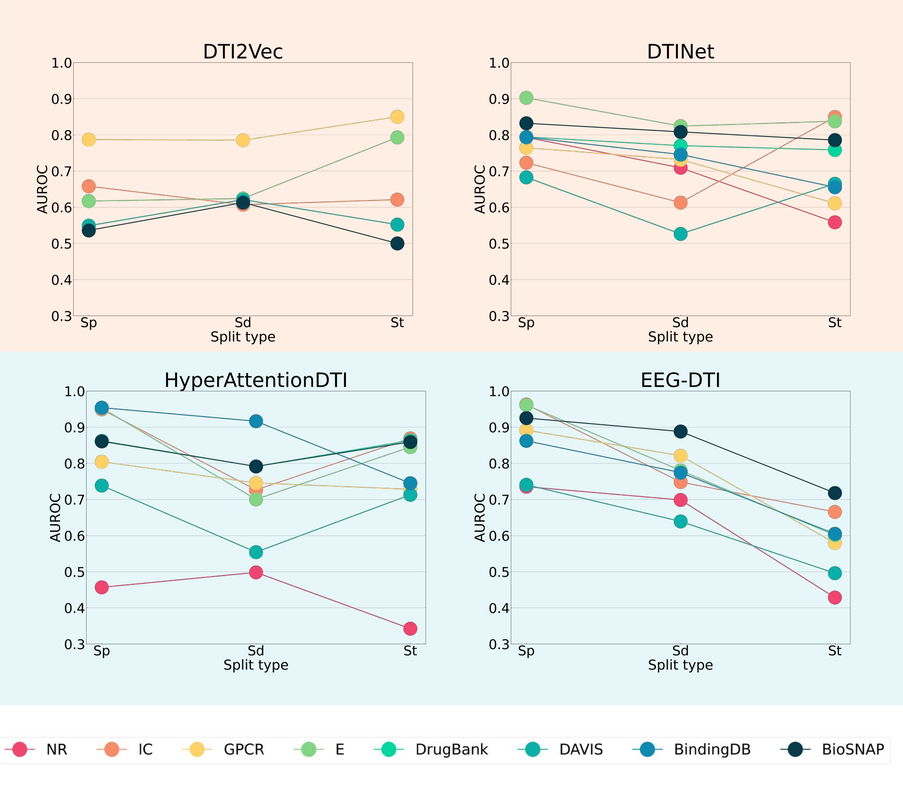
INTEGRATED FRAMEWORK ALLOWS A FAIR COMPARISON



** Protein Characteristics include similarity matrices or kernels. Drug Characteristics include drug similarity, fingerprints and molecular descriptors.*

SPLITTING TECHNIQUES TO PREVENT OVERFITTING

- Results presented are run with 5-times 10-fold cross validation.
- Sd and St accuracies tend to be smaller than Sp.
- The different results in Sd and St for different datasets may be related to the number of drug and protein nodes.
- In smaller datasets, e.g., Yamanishi NR, the models lose accuracy appreciably when changing the type of splitting.



TESTING THE OVERSIMPLICITY OF RANDOM SUBSAMPLING

Preliminary results:

- No big difference in accuracy between random-generated split and RMSD-based split ($\Delta(AUROC_{rnd}, AUROC_{rmsd}) \leq 0.1$)
- Currently:
  - Performing more test .
  - Considering other metrics.

DISCUSSION

- Most models lack reproducibility standards.
- Collecting complementary matrices is a laborious task and authors tend to not compare their algorithm using the same benchmarks.
- True negatives from affinity experiments are not used, instead negative edges come from random subsampling.
- It seems that RMSD-based subsampling does not give any advantages.
- Using node2vec for subsampling not feasible as it presents data leakage problems.
- The proposed evaluation scheme can also serve as a tool to measure the robustness of the model. St and Sd reduce overfitting.

BIBLIOGRAPHY

Cima Universidad de Navarra · Stanford University · Tecnun Universidad de Navarra · ESCUELA DE INGENIERÍA INGENIARITZA ESKOLA SCHOOL OF ENGINEERING

Contact address: uveleirocar@unav.es